



V8 – External View



M12 7816

ENGINE - V8

Description

General

The V8 petrol engine is a 4.4 litre, eight cylinder water cooled engine having two banks of four cylinders, arranged at 90 degrees to each other, with four valves per cylinder. The engine emissions comply with ECD3 (European Commission Directive) and NAS LEV 2 legislative requirements. The engine comprises five main castings - two cylinder heads, cylinder block, timing cover and the oil sump, all of which are manufactured from aluminium alloy.

Plastic rocker covers are fitted to the cylinder heads and, between the cylinder heads, is a plastic manifold intake assembly. A plastic moulded acoustic cover is fitted over the upper engine to reduce engine generated noise.

The engine is supported, between cast mounting brackets and the front subframe, on two self-contained hydraulically damped mounts.

Technical Features

The technical features include:

- Four valves per cylinder
- Single plenum chamber
- Variable Camshaft Control (VCC) system
- Double roller timing chain
- Hydraulic Valve Adjusters (HVA)
- Water Cooled Oil Cooler.

Engine Data

The technical data is detailed below

Engine type	90° V
Cylinders	8
Displacement	4398 ccm
Stroke/bore	82.7/92.0 mm
Maximum Power	210 kW (285 PS; 282 bhp) @ 5400 rev/min
Maximum Torque	440 Nm (324 lbf.ft) @ 3600 rev/min
Compression ratio	10: 1
Firing order	1-5-4-8-6-3-7-2

The cylinders are numbered as follows:

Left bank	5 — 8
Right bank	1 — 4

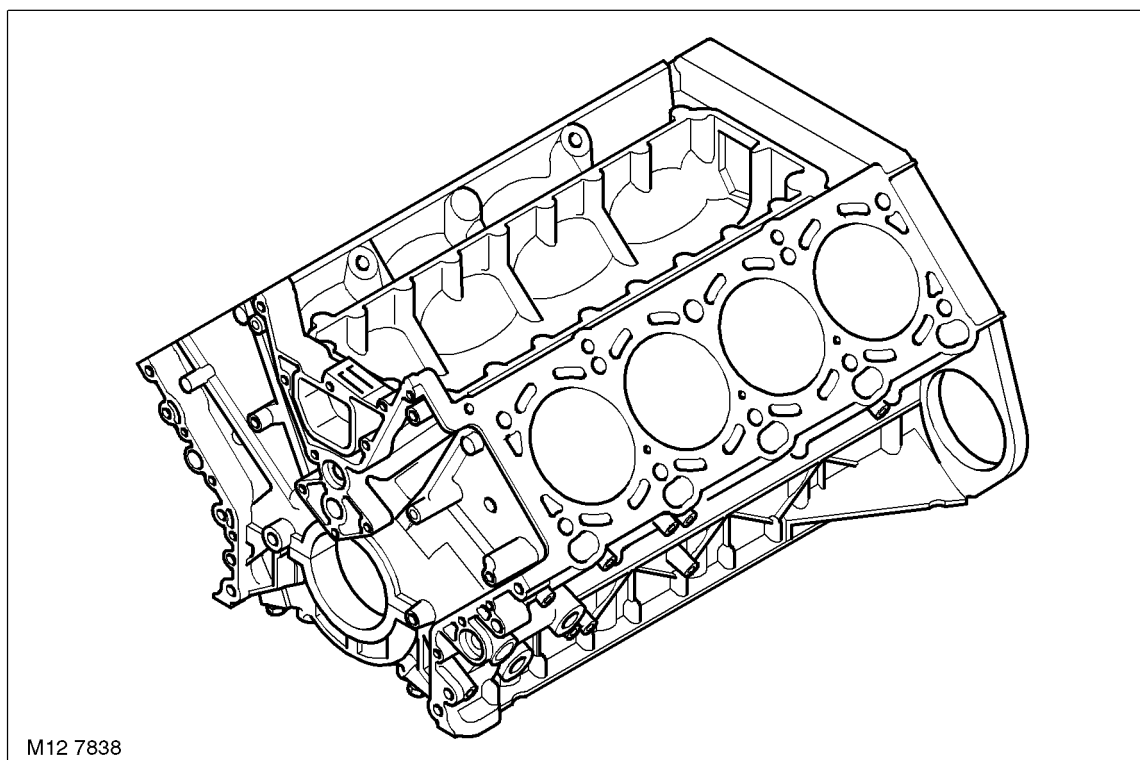
NOTE: Cylinders 4 and 8 are at the rear of the engine

Cylinder Block Components

The cylinder block components are described below:



Crankcase



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The crankcase is made from Alusil for markets where the fuel contains a high sulphur content. The cylinder barrels have a nickel dispersion coating (Nikasil), and are finished by polishing only.

An adaptor plate for the feed and return flexible pipes to the oil filter assembly is fitted to the left side of the crankcase.

A water drain plug is located at the lower LH rear corner to drain coolant from the crankcase.

Crankcase Ventilation

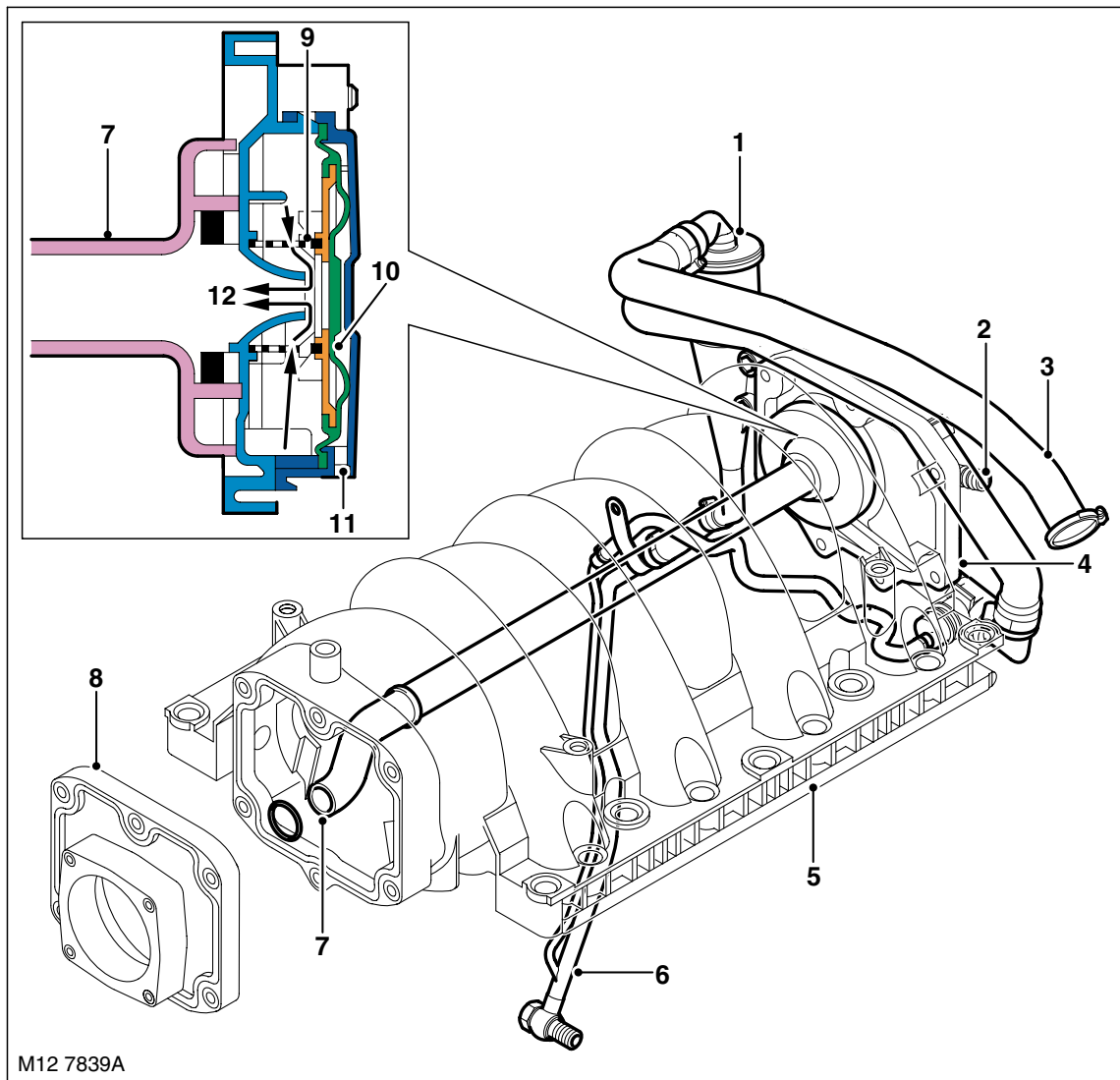
The crankcase ventilation system is pressure regulated by a valve mounted at the rear of the inlet manifold. Vacuum supply ports are incorporated on the valve for the power brake booster and the fuel pressure regulator.

The pressure regulating vent valve varies the vacuum applied to the crankcase depending on engine load. The valve is balanced between spring pressure and the amount of manifold vacuum. At idle when the vacuum is high, the valve closes down and only allows a small amount of blow-by vapours to pass into the manifold. At part to full load conditions, the spring opens the valve and additional blow-by gasses flow into the manifold. An additional tube leads the blow-by gasses through the manifold and up to the mixing plate.

The crankcase ventilation system is completely sealed. Hose clamps are used at all connections to ensure that the system does not contain any leaks.

The crankcase ventilation system uses a cyclone separator to purge the blow-by gasses of any oil vapours. The oil vapours condense in the separator and drain back into the sump.

Crankcase Ventilation System

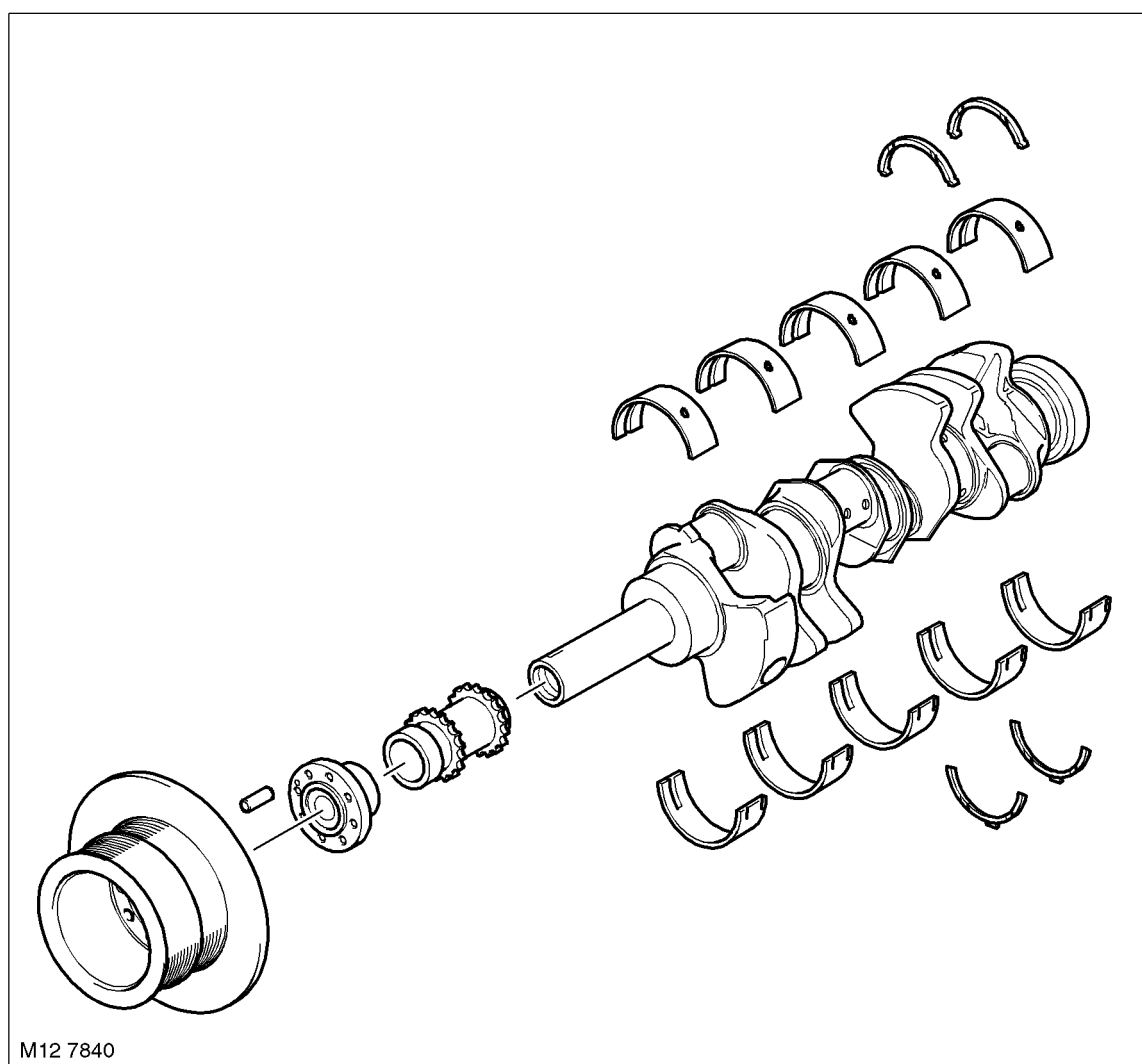


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| 1 Oil separator | 7 Inlet manifold vacuum pipe |
| 2 Breather hose, oil separator to vent valve | 8 Mixing plate |
| 3 Breather hose, LH camshaft cover to oil separator | 9 Vent valve spring |
| 4 Inlet manifold cover and vent valve | 10 Vent valve diaphragm |
| 5 Inlet manifold | 11 Atmospheric vent |
| 6 Oil drain pipe assembly | 12 Blow-by gasses |



Crankshaft



The crankshaft is manufactured from forged steel with five main bearings and the thrust bearing on number five bearing. Six counterweights fitted. These are positioned in harmony with the ignition firing order to ensure smooth running.

The crank pins are offset by 90° and hollow bored to reduce weight.

There are four sizes of crankshaft available, standard and three undersizes.

Colour coding identifies the size of the journal.

The main bearing caps have a unique four bolt fixing for additional strength, and threaded protective bushes which are torqued before the collar screws are installed.

NOTE: For information on colour coding and journal sizes, refer to the General Data section of the workshop manual.

At the front of the crankshaft, incorporated in the auxiliary belt pulley is a torsional vibration damper. Fitted behind the damper, to the crankshaft, is the drive sprocket for the camshaft and oil pump.

Torsional Vibration Damper

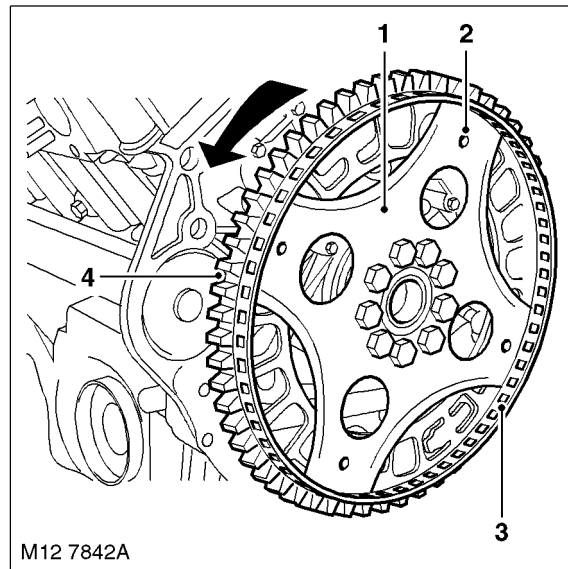
The torsional vibration damper with decoupled belt pulley suppresses longitudinal vibration and reduces noise. It is mounted at front of the crankshaft with 4 bolts torque loaded to 45 Nm.

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Bearing Shells

The upper bearing shells have oil grooves and a drilling to provide oil, via drillings in the crankshaft to the crankshaft main bearings and the connecting rod journals

Flywheel



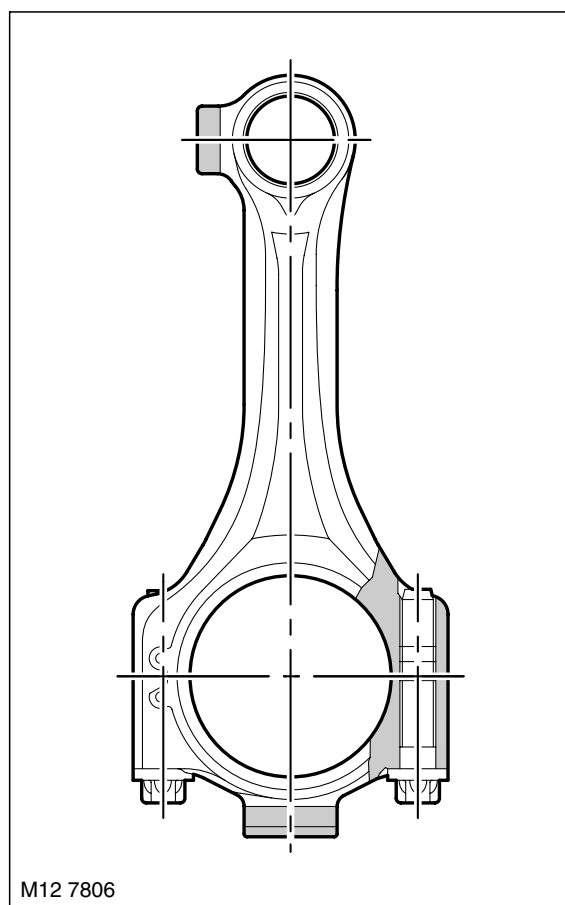
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| 1 Flywheel | 3 Reluctor ring |
| 2 Torque converter attachment holes | 4 Starter motor ring gear |

The single piece steel flywheel is bolted to the rear of the crankshaft with nine bolts. The torque converter is bolted to the rear of the flywheel with four bolts.

Attached to the flywheel is the reluctor ring for the crankshaft position sensor and the starter motor ring gear.



Connecting Rods



The connecting rods are sintered type, one piece forged rods which feature a fracture-split at the big-end between the connecting rod and the bearing cap. The mating surface of the rod and bearing cap form a rough surface (interference fit) that eliminates the need for centering sleeves. Each matched rod and bearing cap pair is stamped with a pairing code.

All of the rods are of one weight class and conform to the double bearing classification, red or blue. One red and one blue bearing shell are fitted to each connecting rod. The red shell is fitted to the bearing cap.

Pistons

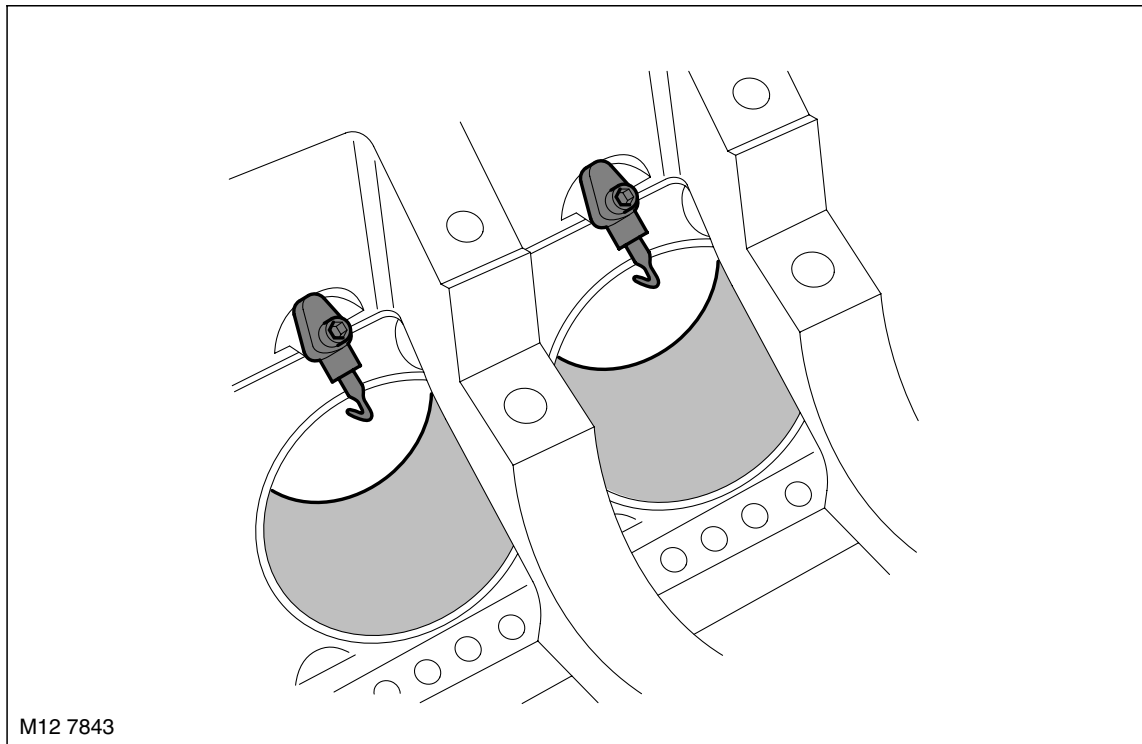
The pistons are of a short skirt design and "iron coated" for use with the Alusil block. Each piston has a flat crown surface with a front facing arrow for installation direction. The relevant cylinder bank number is engraved into the piston crown.

Each piston is fitted with two compression rings and an oil control ring.

There are no thermal control strips in the pistons. The pistons are cooled with specific spray jets, mounted in the crankcase.

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Piston Lubrication Jets

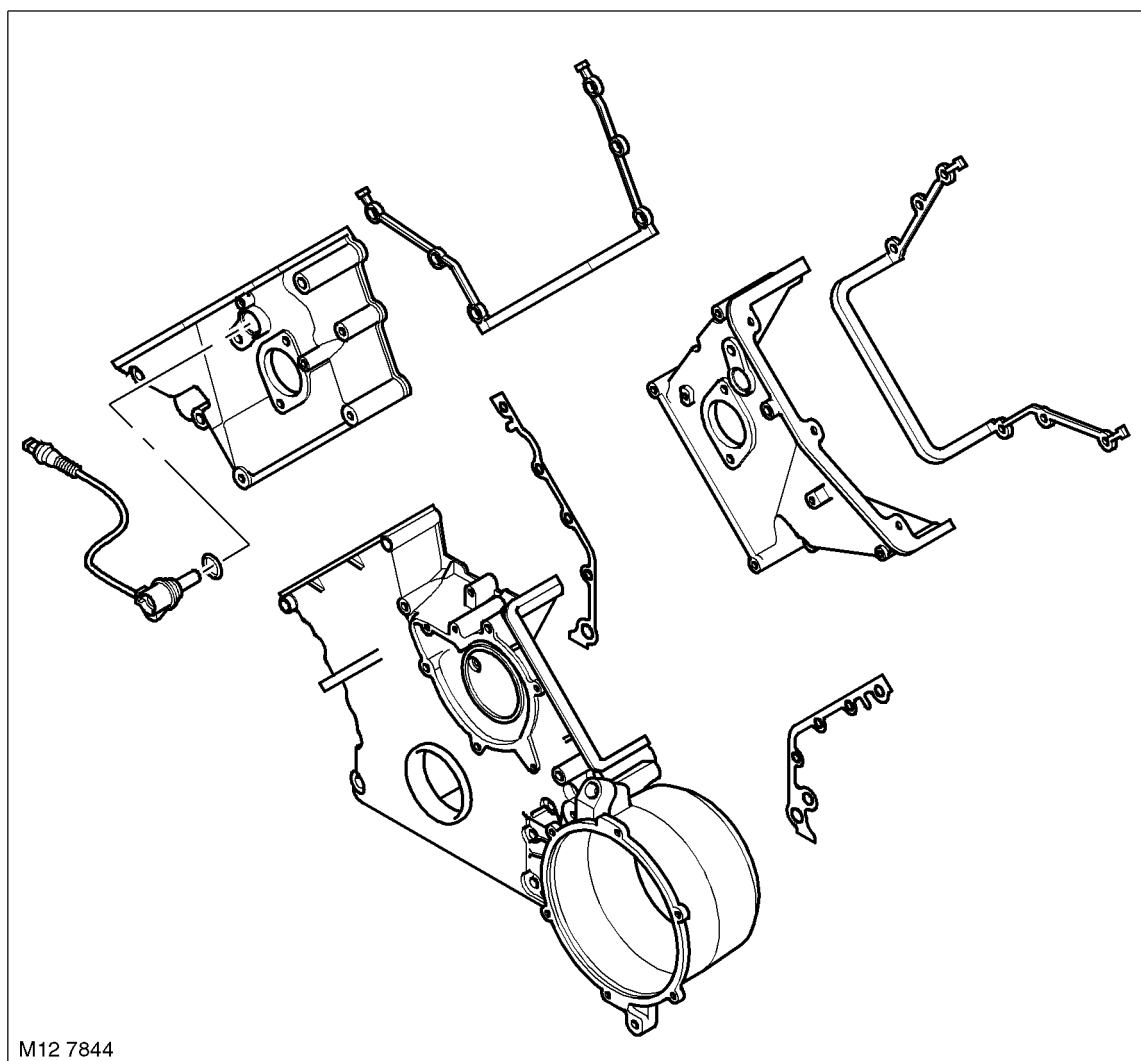


Hook-type nozzles are connected directly into an oil gallery in the crankcase and secured by two screws.

The jets provide lubrication to the cylinder walls, and to the piston undershirt for cooling the pistons and lubricating the gudgeon pins and small-end bearings.



Timing Case Covers

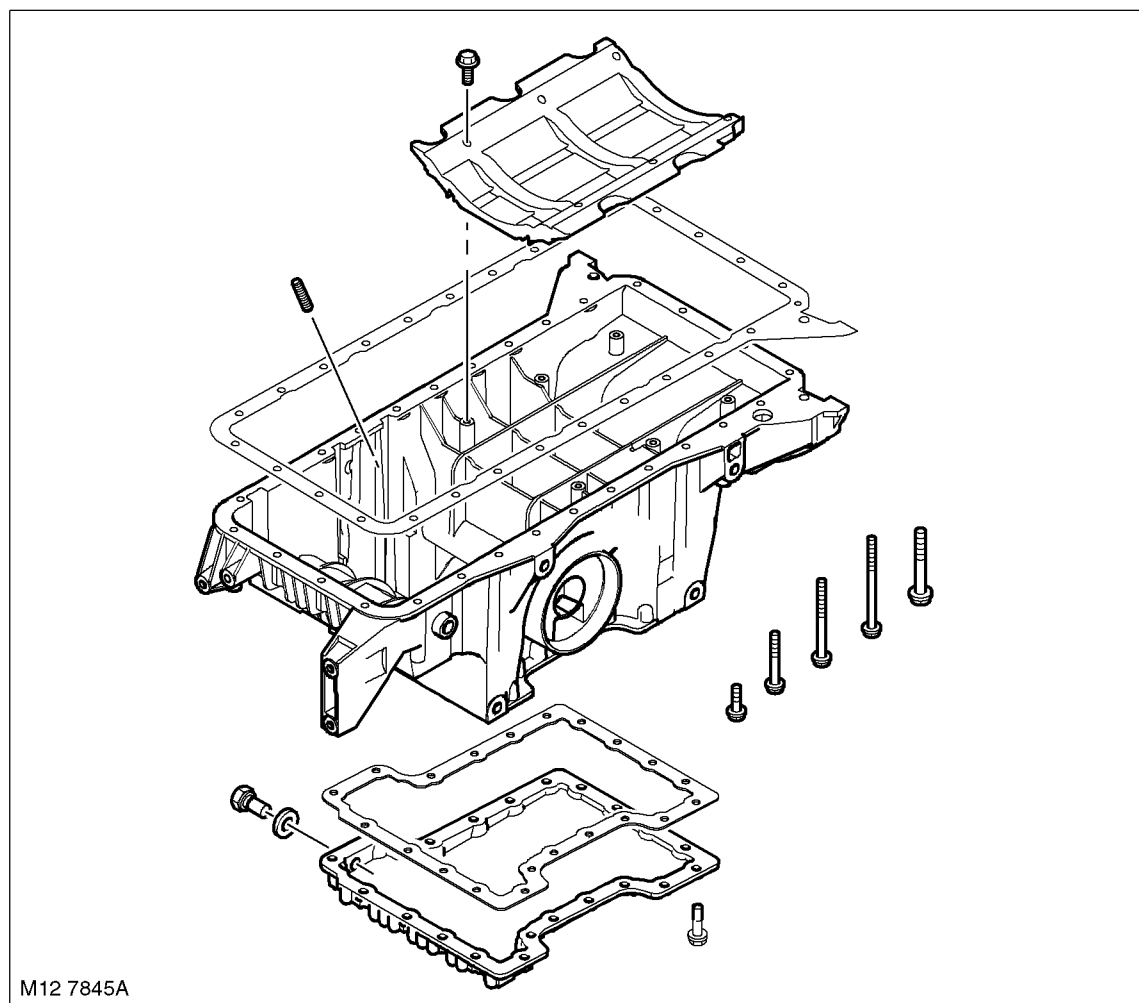


Separate timing case covers are bolted to the two cylinder heads and the crankcase.

The upper timing case covers have ports for the VCC solenoid valves and Camshaft Position (CMP) sensors. The lower cover has an integral mounting for the alternator which acts as a water jacket. A drain plug and sealing ring is fitted to the bottom of the mounting.

Rubber pre-formed gaskets form seals between the cylinder heads and crankcase and respective timing case covers.

Sump



The two-piece aluminium die-cast sump, with an integral tunnel for the differential drive shaft, is sealed to the lower crankcase extension using a rubber metal-backed gasket. The sump is fixed to the lower crankcase extension using 25 bolts. An oil deflector plate is attached to the crankcase reinforcing shell above the sump. The sump incorporates a drain plug and a dip-stick guide tube.

Oil Pump

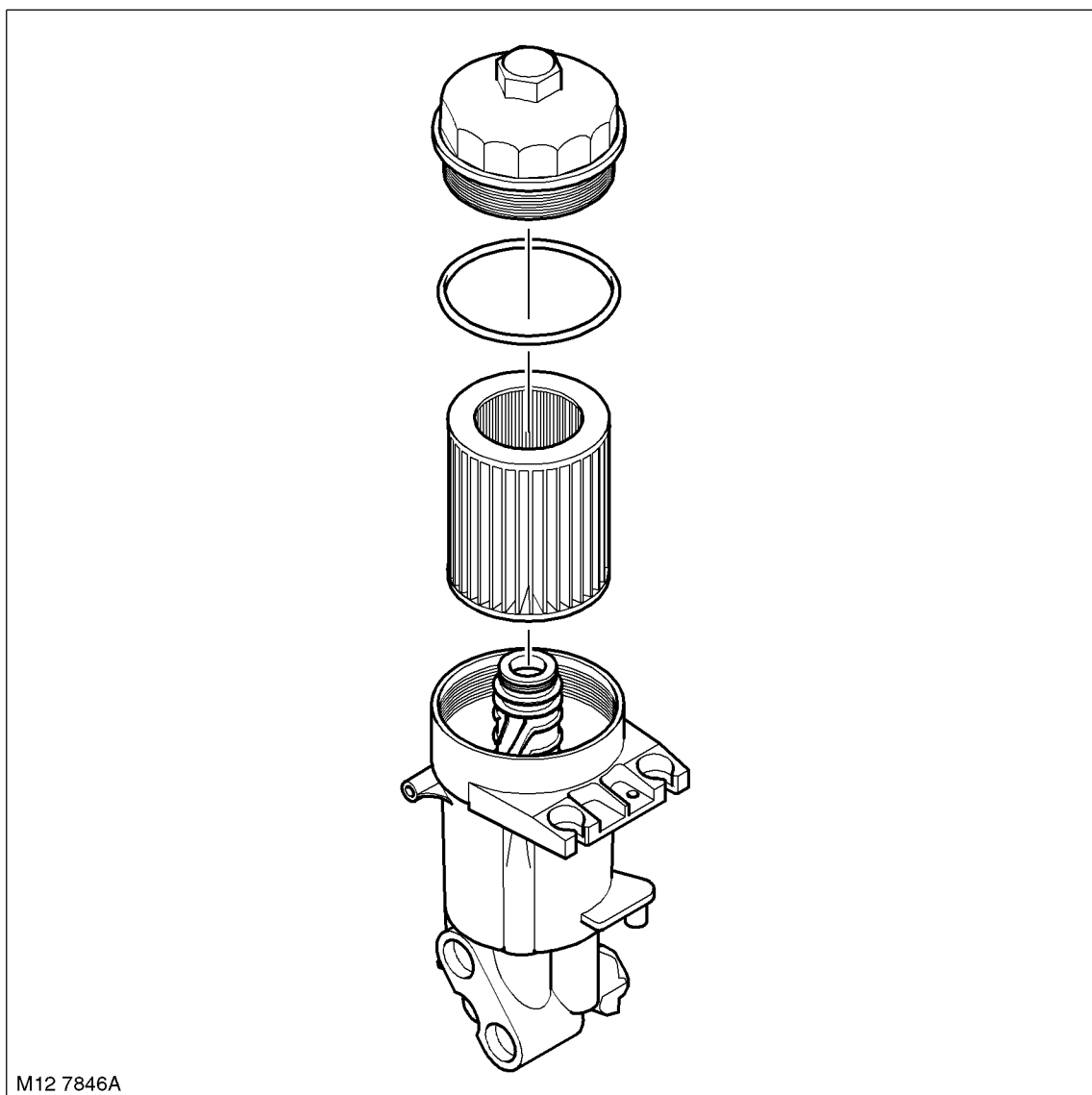
The oil pump housing and cover is manufactured from cast aluminium. The gear type oil pump is driven via the crankshaft.

The oil pump is located in the sump and is attached to Nos 1 and 2 main bearing caps. The rotor-type pump produces a regulated pressure of approximately 4.5 bar.

A metal suction pipe obtains oil from the sump.



Oil Filter Assembly



The oil filter assembly, with a replaceable element, is mounted remote to the engine. It is attached to a mounting bracket fitted to the left of the radiator and the filter is mounted on two rubber mounts to reduce vibration.

Two re-inforced flexible hoses, feed and return, are fitted between the filter assembly and the crankcase mounted adaptor plate. The hose connections are of the push fit type and sealed with an "O" ring seal. The filter hose connections are secured by a locking plate bolted to the filter body. The sump hose connections are secured to the crankcase adaptor with a securing bracket. This bracket is attached with a stud and nut.

A by-pass valve is located internally in the lower end of the filter body. This allows the oil to by-pass the filter if the filter element becomes blocked

A second valve is screwed into the filter body. It is connected to a rigid pipe on the sump by a flexible hose secured by two hose clips. The rigid pipe has a banjo fitting on the sump connection. When the filter is removed, the valve opens allowing oil in the housing to drain to the sump.

The oil pressure switch screws into the base of the filter body.

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Cylinder Head Components

The cylinder head components are described in the following paragraphs.

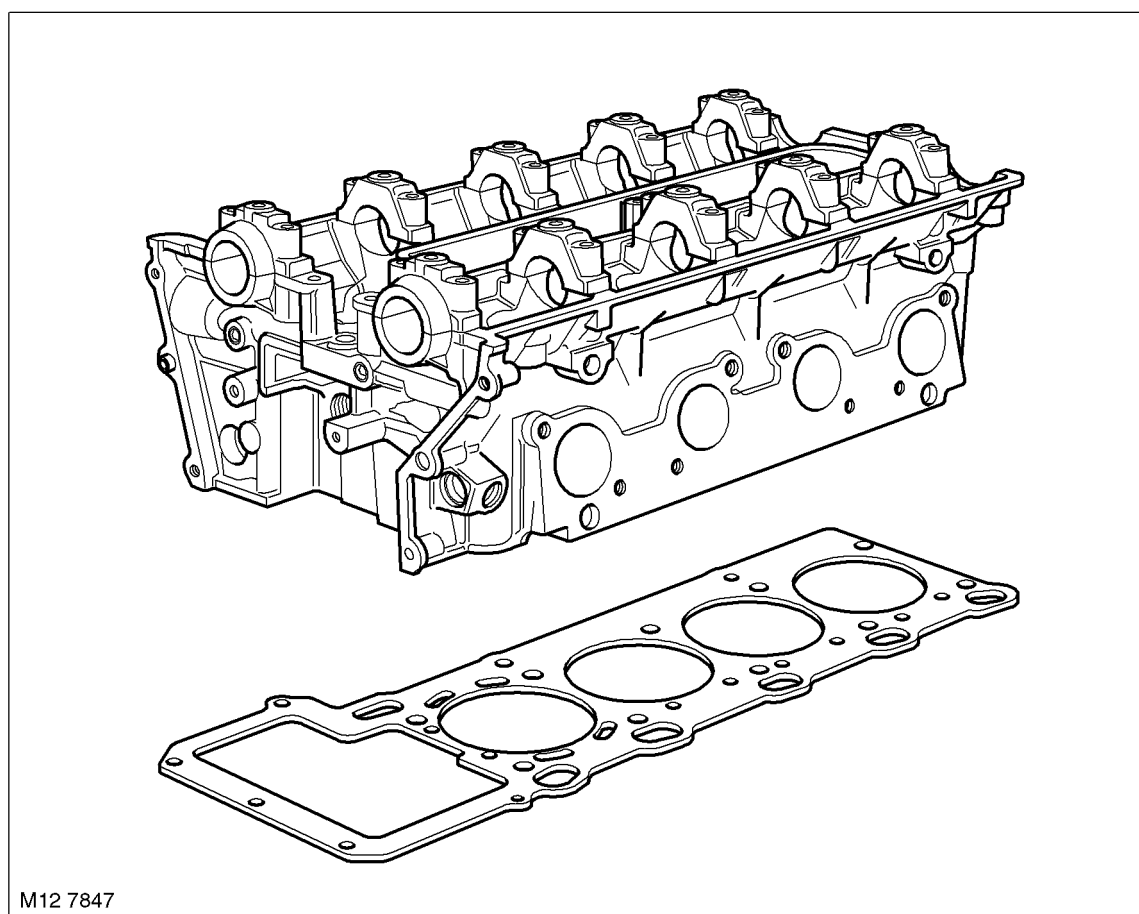
Cylinder Head Cover

The cylinder head cover is manufactured from magnesium alloy and is secured to the cylinder head with 11 threaded spacers with rubber seals. The right hand cover incorporates an oil filler cap.

The cylinder head cover also provides mounting for the coils. The coils are protected by a plastic cover, sealed with a rubber gasket. The cover is retained with two bolts which are covered with protective caps.

NOTE: Damage to the magnesium cover may cause corrosion.

Cylinder Heads



The cross flow cylinder heads are made of die-cast aluminium and feature:

- Centrally located spark plugs
- Four valves per cylinder
- Two camshafts
- Bucket tappets with hydraulic valve clearance adjusters.

The combustion chambers are machined into the heads, thus ensuring more uniform “matched” combustion chambers and compression pressures.

Bolted to the front end of the cylinder head are the oil distribution flanges. The flanges provide a mounting for the VCC solenoids and the advance-retard ports from the solenoids to the intake camshafts.



The cylinder heads for each bank are different lengths due to the offset of the cylinder rows (cylinder bank 1–4 is shorter).

NOTE: The cylinder head gasket part numbers are different for the two cylinder banks.

The cylinder head is machineable to a limit of 0.3 mm. A repair cylinder head gasket is available and is identified by a small hole beside the 4th and 8th cylinder apertures.

Camshafts

Two solid cast iron camshafts are used for each cylinder head. The cams run in five machined bearing surfaces with numbered caps for identification.

The camshafts are also marked for identification and incorporate hexagonal machined blocks for timing the cams during installation.

For installation and servicing, the camshafts are held in position with locking pins. These pins are inserted into the front camshaft bearing caps on both cylinder heads.

Valves

The intake and exhaust valves are arranged in a “V” pattern. The valves use single, conical, coil bound shaped valve springs and Hydraulic Valve Adjustment (HVA) tappets and each comprises the following components:

- Valve x 1 (inlet or exhaust)
- Valve stem seal x 1
- Valve spring x 1
- Valve spring seat x 1
- Valve spring cap x 1
- Valve collets x 2
- Tappet x 1.

Hydraulic Valve Adjustment Tappets

Bucket type hydraulic tappets are used for adjusting the valve clearance. The buckets are self-breathing with a carbon nitride finished cam contact surface.

Primary drive

Primary drive is provided by a single roller chain from the crankshaft to the two rubber coated cam gears. The train is guided by a centrally positioned V-shaped aluminium deflector rail, a straight rail on cylinder bank 5 to 8 and a curved tensioner rail on cylinder bank 1 to 4.

Timing chain adjustment is carried out with a hydraulic tensioner mounted on the upper case for cylinders 1 to 4

Secondary Drive

Secondary drive is from the intake camshaft to the exhaust camshaft by a single roller chain. Chain tension is maintained by a hydraulic, spring assisted tensioner that incorporates an oil spray jet.

The oil supply for the chain tensioner also supplies the VCC hydraulic units through a pressure control valve.

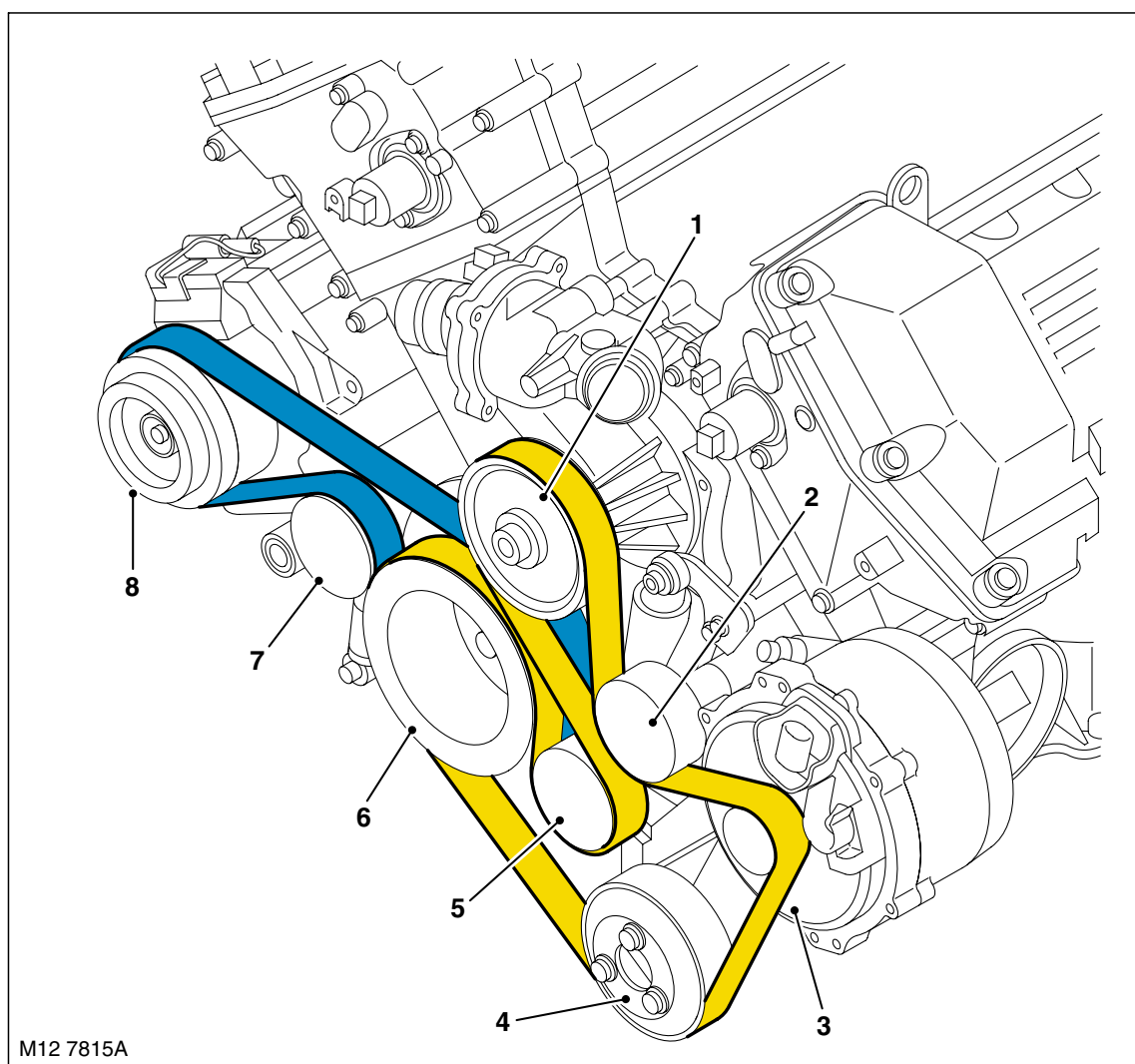
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Ancillary Components and Belt Drive

The ancillary components, the torsional vibration damper, alternator, A/C compressor, steering pump and water pump, are driven by the engine crankshaft by the aid of the ancillary drive belts. The A/C compressor is driven by a separate belt.

The belts, which are maintenance free poly-vee type belts, are automatically pre-loaded by two tensioning spring pots and are routed over deflection pulleys in order to maintain sufficient adhesion about the ancillary pulleys. This ensures slip-free drive of the ancillary components.

Belt Drive



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| 1 Coolant pump | 5 Tensioner pulley |
| 2 Idler pulley | 6 Crankshaft pulley |
| 3 Alternator | 7 Tensioner pulley |
| 4 Power assisted steering pump | 8 A/C compressor |

Air Intake System

The inlet manifold is a light weight, one piece plastic moulding. Fuel is supplied by a single fuel rail and regulated by a single pressure regulator mounted on the end of the rail. The manifold is acoustically decoupled from the cylinder heads to reduce noise and vibrations.

The throttle plate has two wedges screwed directly to it, these provide a curved zone for smoother throttle response during idle, off-idle transition.

The combined output of the mixing plate ensures that the gasses and vapours are evenly distributed among all the cylinders, which can improve idle quality.



Lubrication

The lubrication system is a wet sump, pressure fed type. It lubricates the engine sliding surfaces, dissipates heat and absorbs fuel combustion residue.

Operation

Oil from the sump is drawn up through a fabricated metal pick-up pipe which contains a mesh to filter out any relatively large particles of material which could cause damage to the oil pump. The head of the pick-up is centrally immersed in the sump oil which is delivered to the rotor pump.

Variable Camshaft Control (VCC)

The V8 engine uses a VCC (Variable Camshaft Control) system for valve timing on the intake camshaft. While the engine is running the inlet camshafts are continuously adjusted to their optimum position.

Both camshafts are simultaneously adjusted to a maximum of 20° of the crankshaft rotational axis. This equates to a maximum span of 40° crankshaft rotation.

 **ENGINE MANAGEMENT SYSTEM – V8, DESCRIPTION AND OPERATION, VCC System.**

Cylinder Bank Components

Each cylinder bank contains the following:

- Cylinder head with oil ports for the VCC system
- Control solenoid valve
- VCC Transmission and sprockets
- Oil distribution flanges
- Oil check valve
- Camshaft position impulse wheels
- Camshaft position sensors.

Control Solenoid Valve

The control solenoid valve is fitted with an integral non-return valve and has four ports:

- 1 Input supply port — engine oil pressure
- 2 Output retard port — to rear of piston/helical gear (retarded camshaft position)
- 3 Output advance port — to rear of piston/helical gear (advanced camshaft position)
- 4 Vent — released oil pressure.

The non-return valve is fitted to the rear of the solenoid valve in the cylinder head oil gallery. The non-return valve maintains an oil supply in the VCC transmission and oil circuits after the engine is switched off. This prevents piston movement when the engine is restarted.

 **ENGINE MANAGEMENT SYSTEM – V8, DESCRIPTION AND OPERATION, VCC System.**

VCC Transmission and Sprockets

The primary and secondary timing chain sprockets are integrated with the self-contained VCC transmission.

The controlled adjustment of the camshaft occurs inside the “transmission”. Controlled oil pressure moves the piston axially.

The helical gear cut of the piston acts on the helical gears on the inside surface of the transmission and rotates the camshaft to the specific advanced or retarded angle position.

Three electrical pin contacts are located on the front surface to verify the default maximum retard position using an ohmmeter. This is required during assembly and adjustment.

Oil Distribution Flanges

The oil distribution flanges are bolted to the front surface of each cylinder head. They provide a mounting location for the VCC solenoids as well as the advance-retard oil ports from the solenoids to the intake camshafts.

Camshafts

Each intake camshaft has two oil ports separated by three sealing rings on their forward ends. The ports direct pressurized oil from the oil distribution flange to the inner workings of the VCC transmission. Each camshaft has left-hand threaded bores in their centres for the attachment of the timing chain sprockets on the exhaust cams and the VCC transmissions for each intake camshaft.

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Camshaft Position Impulse Wheels

The camshaft position impulse wheels provide camshaft position status to the engine control module via the camshaft position sensors. The asymmetrical placement of the sensor wheel pulse plates provides the engine control module with cylinder specific position ID in conjunction with crankshaft position.

 **ENGINE MANAGEMENT SYSTEM – V8, DESCRIPTION AND OPERATION, Bosch ME 7.2 Engine Management System.**

Camshaft Position Sensors

There are two CMP position sensors which are located on the upper timing case covers. The camshaft position sensors monitor the position of the camshafts to establish ignition timing order, fuel injection triggering and for accurate Variable Camshaft Control (VCC) camshaft advance-retard timing feedback. The camshaft position sensor is a Hall-effect sensor which switches a battery fed supply on and off. The supply is switched when the teeth machined onto the camshaft gear pass by the tip of the sensor. The four teeth are of differing shapes, so the ECM can determine the exact position of the camshaft at any time.

 **ENGINE MANAGEMENT SYSTEM – V8, DESCRIPTION AND OPERATION, Bosch ME 7.2 Engine Management System.**



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